Class 4 – Pituitary Tumor: Patient & Caregiver Guide

An educational guide for people whose brain MRI suggests a pituitary tumor — a growth in the pituitary gland, the small hormone-control gland at the base of the brain. It explains what a pituitary tumor is, its types, symptoms, diagnosis, MRI features, treatment (including hormone therapy), recovery, nutrition, rehabilitation, warning signs, outlook, and answers to common questions.

Educational disclaimer: This document is for general education and for use in a retrieval-augmented (RAG) AI assistant that supports patients after an MRI-based brain scan. It is NOT a medical diagnosis and does NOT replace consultation with qualified healthcare professionals such as a neurologist, neurosurgeon, or oncologist. Every person is different, and only your own medical team can confirm a diagnosis and plan treatment. If you have severe or sudden symptoms, seek emergency medical care immediately.

1. Overview

A pituitary tumor is a growth in the pituitary gland, a pea-sized gland at the base of the brain that controls many of the body's hormones. Most pituitary tumors are called pituitary adenomas. The great majority are benign (non-cancerous) and slow-growing; cancerous pituitary tumors are very rare.

Two key ideas: (1) Functioning tumors make too much of a hormone, causing specific conditions such as acromegaly (excess growth hormone), Cushing's disease (excess cortisol), or high prolactin (a prolactinoma). (2) Non-functioning tumors do not over-produce hormones and usually cause problems by their size, pressing on nearby structures such as the vision nerves.

Where it occurs: in the pituitary gland, which sits in a small bony pocket (the sella turcica) just below the brain and behind the eyes, close to the optic nerves. Epidemiology: pituitary tumors are common and are often found by chance. Size: 'microadenomas' are under 10 mm; 'macroadenomas' are 10 mm or larger. Age groups: they can occur at any age, most often in adults. Gender: prolactinomas are more common in younger women.

Disease progression and prognosis: most pituitary tumors grow slowly and the outlook is generally very good. Many are controlled with medicine or cured with surgery. Hormone effects and vision problems often improve with treatment.

2. Brain Anatomy

The pituitary gland is small but powerful, and its position explains many symptoms.

- Pituitary gland: the 'master gland' that controls other hormone glands, including the thyroid, adrenal glands, and reproductive organs.
- Hypothalamus: the brain area just above that directs the pituitary; tumors can affect this link.
- Sella turcica: the bony pocket that holds the gland.
- Optic chiasm: the crossing point of the vision nerves sits just above the gland; a growing tumor can press here and affect side vision.
- Cavernous sinuses: spaces on each side carrying blood and the nerves that move the eyes; large tumors can extend into them.
- Key hormones: growth hormone, prolactin, ACTH (controls cortisol), TSH (controls thyroid), and the hormones that control the ovaries and testes.

Relationship to surrounding tissue: because the gland sits right below the optic chiasm, a larger tumor often first affects vision (classically the outer/side fields). Its position also allows surgeons to reach it through the nose (transsphenoidal surgery) rather than opening the skull.

3. What Causes a Pituitary Tumor?

In most people the cause is unknown. Pituitary tumors arise when cells in the gland grow abnormally. They are not caused by anything the patient did and are not contagious.

Established Risk Factors

- Rare inherited syndromes: such as Multiple Endocrine Neoplasia type 1 (MEN1), Carney complex, and familial isolated pituitary adenoma (FIPA).
- Family history of pituitary tumors in a small number of cases.

Possible / Being Studied Factors

- Hormonal changes may influence some tumors, but a clear avoidable cause is usually not found.

Protective / Not Proven Factors

- No specific action is proven to prevent pituitary tumors.
- A healthy lifestyle supports general health but is not proven to prevent them.

4. Types and Classification

Pituitary tumors are usually classified by size and by whether they over-produce a hormone, rather than by the WHO I–IV grading used for gliomas. Most are benign adenomas. The rare pituitary carcinoma is the only truly malignant form.

Type	What it is	Typical effect	Typical treatment
Prolactinoma	Makes too much prolactin.	Irregular periods, milk production, low libido, infertility.	Usually medicine (dopamine agonists) first
Growth-hormone tumor	Makes too much growth hormone.	Acromegaly (enlarged hands, feet, face) in adults.	Surgery, sometimes medicine or radiation
ACTH tumor	Makes too much ACTH, raising cortisol.	Cushing's disease (weight gain, high blood pressure, high sugar).	Surgery, sometimes medicine or radiation
Non-functioning adenoma	Does not over-produce hormones.	Pressure effects: vision loss, headache, low hormone levels.	Surgery if causing symptoms; sometimes observation
Micro- vs macroadenoma	Under 10 mm vs 10 mm or larger.	Larger tumors more likely to press on vision nerves.	Depends on size, hormones, and symptoms

Grade and recurrence: most adenomas are benign and slow-growing with a low recurrence risk after good treatment. Aggressive adenomas and the very rare carcinoma need closer, specialist follow-up.

5. Common Symptoms

Symptoms come from two sources: too much of a hormone (functioning tumors) and pressure from a larger tumor (especially on the vision nerves). Some tumors also cause too little hormone by damaging the normal gland.

- Vision changes: loss of side (peripheral) vision or blurred/double vision, from pressure on the optic nerves — a classic sign of a macroadenoma.
- Headache: from stretching of the tissues around the gland.
- Hormonal imbalance: irregular or absent periods, milk production (galactorrhoea), low libido, or infertility (often from high prolactin).
- Acromegaly signs: enlarging hands, feet, and facial features, and joint pain (excess growth hormone).
- Cushing's signs: weight gain around the middle and face, easy bruising, high blood pressure, and high blood sugar (excess cortisol).
- Fatigue and weakness: from low hormone levels when the tumor presses on the gland.
- Feeling cold, weight change, and low mood: from an underactive thyroid or adrenal axis.
- Nausea and vomiting: with larger tumors or sudden changes.

> Pituitary emergency (apoplexy): a sudden severe headache with sudden vision loss, double vision, vomiting, or collapse can mean bleeding into the tumor. This is a medical emergency — seek immediate care.

6. Diagnosis

Diagnosing a pituitary tumor combines imaging with detailed hormone testing, because both the tumor's size and its hormone effects matter.

- Medical history: symptoms of hormone excess or deficiency, vision changes, and family history.
- Neurological and vision examination: including formal visual field testing to check side vision.
- MRI: the main imaging test; a dedicated pituitary MRI shows the gland and tumor in detail without radiation.
- Contrast MRI: dye helps show the tumor against the normal gland; adenomas often enhance less than the surrounding gland.
- CT scan: used in emergencies or to show bony detail; less detailed than MRI for the gland.
- Hormone (blood) tests: measure prolactin, growth hormone/IGF-1, cortisol/ACTH, thyroid hormones, and reproductive hormones to find over- or under-production.
- Special hormone tests: such as suppression or stimulation tests for Cushing's or acromegaly.
- Genetic testing: considered if an inherited syndrome (like MEN1) is suspected.
- Biopsy: not usually needed separately; tissue is examined after surgery.

Advantages and limitations: MRI shows the tumor well, but hormone tests are essential to understand its type and effects and to plan treatment. Vision testing is key when the tumor is large.

7. MRI Characteristics

A dedicated pituitary MRI shows the size, position, and spread of the tumor.

- Location: within the pituitary gland in the sella turcica, at the base of the brain.

- Size: microadenoma (under 10 mm) or macroadenoma (10 mm or larger).
- Shape: microadenomas are small spots within the gland; macroadenomas are larger and may bulge upward.
- Contrast enhancement: microadenomas often enhance less than the normal gland early after dye.
- Suprasellar extension: larger tumors grow upward and can press on the optic chiasm, sometimes giving a 'snowman' or 'figure-of-eight' shape.
- Cavernous sinus involvement: the tumor may extend sideways into these spaces.
- Edema: significant surrounding brain swelling is uncommon.
- Haemorrhage or necrosis: bleeding into the tumor (apoplexy) can be seen and is an emergency.

A typical macroadenoma looks like a mass rising out of the sella that lifts or presses the optic chiasm, which explains the classic loss of side vision.

8. Treatment Options

Treatment depends on the tumor type (which hormone, if any, it makes), its size, and its effect on vision. Pituitary tumors are unusual in that medicine alone can cure or control some of them.

Observation (Watchful Waiting)

Small, non-functioning tumors that cause no symptoms and do not threaten vision are often simply monitored with periodic MRI and hormone tests.

Surgery

- Transsphenoidal surgery: the main operation, done through the nose and sphenoid sinus to reach the gland without opening the skull; recovery is usually quicker than a craniotomy.
- Endoscopic surgery: a camera-guided version of transsphenoidal surgery, widely used.
- Craniotomy: rarely needed, for very large or complex tumors that cannot be reached through the nose.
- Complete or partial removal: surgeons remove as much tumor as is safe, especially to relieve pressure on the vision nerves.
- Benefits: can cure many tumors, restore vision, and correct hormone excess.
- Risks: nasal issues, leak of brain fluid (CSF), infection, and temporary or lasting hormone changes needing replacement.
- Recovery: often a short hospital stay with return to light activity over a few weeks.

Radiation Therapy

- Stereotactic radiosurgery (Gamma Knife, CyberKnife) and external beam radiation: used for tumor that remains or regrows after surgery, or when surgery is not suitable.
- Side effects: fatigue and, over time, possible reduced pituitary function needing hormone replacement.
- Recovery: hormone effects are monitored for years after radiation.

Chemotherapy

Standard chemotherapy is not used for typical pituitary adenomas. A drug called temozolomide is reserved for the rare aggressive adenoma or carcinoma, under specialist care.

Targeted Therapy and Immunotherapy

These are not standard treatments and are studied mainly for rare, aggressive tumors within clinical trials.

Hormone Therapy (Especially Important for Pituitary Tumors)

- Dopamine agonists (e.g., cabergoline, bromocriptine): often the first treatment for prolactinomas, frequently shrinking the tumor and normalising prolactin without surgery.
- Somatostatin analogues and other drugs: used to control growth-hormone tumors (acromegaly) and some others.
- Hormone replacement: if the tumor or its treatment leaves the gland underactive, patients may need thyroid hormone, steroids (hydrocortisone), sex hormones, or growth hormone to replace what is missing.
- Careful monitoring: hormone levels are checked regularly to balance replacement and avoid too much or too little.

> Steroid (cortisol) replacement can be life-saving. If you are told you need steroid replacement, never stop it suddenly, and follow 'sick day rules' to increase the dose during illness, as advised by your endocrinologist.

9. Recovery Timeline

Recovery is often quicker than for other brain tumors, especially after transsphenoidal (through-the-nose) surgery. These are general guides.

Pathway	What to expect
After transsphenoidal surgery	Hospital stay often 2–4 days; nasal congestion for a few weeks; light activity in 1–2 weeks; fuller recovery over 4–6 weeks.
After radiation/radiosurgery	Often outpatient; fatigue is mild; hormone effects are watched for years.
Medicine-controlled (e.g., prolactinoma)	No surgical recovery; the tumor may shrink and symptoms improve over weeks to months on medication.
Observation	No treatment recovery; periodic MRI and hormone tests.

- Return to work: often 1–4 weeks after surgery for desk jobs; longer for heavy work.
- Return to school: gradual return once recovered and cleared.
- Driving: usually paused after surgery or if vision is affected; resume after clearance and normal vision testing.
- Physical activity: avoid nose-blowing, heavy lifting, and straining early after transsphenoidal surgery to protect the healing area; gentle activity is encouraged.

Expected milestones: vision often improves within days to weeks after pressure is relieved; hormone levels and energy stabilise over weeks to months with the right treatment and replacement.

10. Home Care

Home care focuses on medication, protecting the surgical area, and watching hormone balance.

- Medication adherence: take hormone medicines and any replacements exactly as prescribed; never stop steroid replacement suddenly.
- After nasal surgery: avoid forceful nose-blowing, heavy lifting, and straining for a few weeks; sneeze with your mouth open.
- Watch fluid balance: report unusual thirst and very frequent urination, which can signal a temporary hormone imbalance (diabetes insipidus) after surgery.
- Sleep and hydration: keep regular sleep and drink water as advised.
- Stress reduction: manage stress; remember sick-day steroid rules if you need replacement.
- Activity restrictions: avoid driving until cleared and vision is normal.
- Mental health: seek support for mood changes, which can accompany hormone shifts.
- Follow-up appointments: keep all scans, vision checks, and hormone blood tests.
- Carry information: if on steroid replacement, carry a steroid alert card or bracelet.

11. Nutrition

Good nutrition supports recovery from pituitary surgery or radiation and hormone balance, helps the body cope with treatment, and keeps energy and mood steady. There is no single 'anti-tumor' diet, but a balanced, mostly whole-food diet is widely recommended. Always follow any specific advice from your medical team or dietitian, especially during treatment.

Key Nutrients and Why They Help

Nutrient	Why it helps	Vegetarian sources	Non-vegetarian sources
Protein	Repairs tissue and supports healing after surgery or treatment.	Dal, beans, chickpeas, rajma, soya, paneer, tofu, nuts	Eggs, chicken, fish, lean meat
Omega-3 fats	Support brain-cell health and reduce inflammation.	Flaxseed, chia, walnuts, soybean oil	Fatty fish (salmon, mackerel, sardines)
Vitamin D	Supports bone and immune health; many patients are low.	Sunlight, fortified milk/cereals, mushrooms	Egg yolk, fatty fish
Vitamin B12	Needed for healthy nerves and blood.	Fortified foods, dairy, supplements if vegan	Eggs, dairy, fish, meat
Vitamin C	Helps wound healing and immunity.	Amla, orange, guava, lemon, capsicum, tomato	(mainly plant foods)
Vitamin E	An antioxidant that protects cells.	Nuts, seeds, vegetable oils, spinach	(mainly plant foods)
Antioxidants	Protect cells from damage; from colourful produce.	Berries, leafy greens, carrots, beetroot, tomatoes	(mainly plant foods)
Healthy fats	Provide steady energy and support the brain.	Olive oil, mustard oil, nuts, avocado, seeds	Fish

Nutrient	Why it helps	Vegetarian sources	Non-vegetarian sources
Whole grains	Give steady energy and fibre for digestion.	Oats, brown rice, millets (jowar, bajra, ragi), whole wheat	(mainly plant foods)
Hydration (water)	Prevents tiredness, headaches, and constipation.	Water, buttermilk, coconut water, soups	Water, broths

Sample Daily Meal Suggestions

- Breakfast: vegetable oats or besan chilla or eggs with whole-wheat toast and a fruit.
- Mid-morning: a fruit (apple, guava, orange) or a handful of nuts.
- Lunch: 2 multigrain rotis, dal or beans, a large bowl of vegetables, curd, and salad.
- Evening: buttermilk, sprouts, or green tea with roasted chana.
- Dinner: brown rice or roti with vegetables and a protein (fish, chicken, tofu, or dal).
- Through the day: sip water regularly to stay well hydrated.

> During chemotherapy or if you have swallowing, taste, or appetite problems, ask your team for tailored advice. Do not start high-dose supplements on your own, as some can interfere with treatment.

12. Foods to Limit

Some foods and habits do not directly cause brain tumors but can slow recovery, worsen general health, or interact with treatment. It is wise to limit the following.

- Alcohol: can interact with medicines (including anti-seizure and chemotherapy drugs), affect the brain, and disturb sleep. Best avoided or strictly limited.
- Smoking and tobacco: harm blood vessels and overall health and slow healing. Stopping is strongly recommended.
- Highly processed foods: often high in salt, sugar, and unhealthy fat and low in nutrients.
- Sugary beverages: sodas and sweetened juices add empty calories and can worsen energy swings.
- Excess salt: too much salt can raise blood pressure and cause fluid retention.
- Excess caffeine: large amounts of coffee, tea, or energy drinks can worsen anxiety, tremor, and sleep problems.
- Trans fats: found in some fried and bakery items; bad for heart and blood vessels.
- Ultra-processed foods: packaged snacks and ready meals; keep these occasional.

In short, a fresh, home-cooked, balanced diet with less salt, sugar, alcohol, and no smoking supports recovery and overall brain health.

13. Rehabilitation

Many people need little physical rehabilitation after pituitary treatment, but support for vision, energy, and hormone adjustment is often helpful.

- Vision therapy and monitoring: to support recovery and adaptation if vision was affected.
- Physical therapy: general reconditioning if weakness or long illness reduced fitness.
- Occupational therapy: help returning to daily tasks and work.
- Cognitive support: for concentration or memory changes linked to hormone imbalance.
- Endocrine rehabilitation: careful, ongoing adjustment of hormone replacement to restore energy and wellbeing.
- Counselling: to cope with body-image changes (e.g., from acromegaly or Cushing's) and the emotional impact of hormone changes.

Duration and goals: adjustment of hormones and recovery of vision and energy may take weeks to months. The goal is to restore normal hormone balance, protect vision, and return to daily life. Outcomes are generally good.

14. Lifestyle Recommendations

Healthy daily habits support brain health, mood, and recovery. Adapt these to your abilities and your doctor's advice.

- Regular exercise: gentle activity such as walking, as approved by your doctor, improves energy, mood, and sleep.
- Stress management: reduce stress with breathing exercises, hobbies, and pacing your day.
- Meditation: mindfulness or guided relaxation can ease anxiety and improve focus.
- Yoga: gentle yoga (with medical clearance) supports flexibility, balance, and calm.
- Sleep hygiene: keep a regular sleep schedule, a dark quiet room, and limit screens before bed; aim for 7–9 hours.
- Hydration: drink water through the day to avoid tiredness and headaches.
- Weight management: keep a healthy weight with balanced food and activity.
- Mental wellness: seek support for low mood or anxiety; counselling can help.
- Family support: involve family in appointments, medication, and daily routines.
- Social engagement: stay connected with friends and community to protect mood and thinking skills.

15. Possible Complications

Complications can come from the tumor, from surgery, or from hormone imbalance.

- Vision loss: from pressure on the optic nerves; often improves with treatment but can be lasting if severe.
- Hormonal imbalance: too much or too little of pituitary hormones, needing medicine or replacement.
- Hypopituitarism: an underactive gland after treatment, requiring hormone replacement.
- Diabetes insipidus: a temporary (sometimes lasting) problem after surgery causing excess urination and thirst, treated with medication.
- CSF leak: leakage of brain fluid through the nose after transsphenoidal surgery, needing prompt treatment.

- Pituitary apoplexy: sudden bleeding into the tumor — a medical emergency.
- Infection or meningitis: a small risk after surgery.
- Recurrence: the tumor can regrow, so monitoring continues.
- Mood disorders: hormone changes can affect mood; these are treatable.

16. Emergency Warning Signs

Seek emergency medical help right away for any of these:

- A sudden, severe ('worst ever') headache, especially with vision loss or double vision (possible pituitary apoplexy).
- Sudden loss of vision or rapidly worsening vision.
- Severe vomiting with headache and confusion.
- Loss of consciousness or collapse.
- Signs of adrenal crisis if on steroid replacement: severe weakness, vomiting, low blood pressure, and confusion — take emergency steroids if advised and get help.
- Clear watery fluid draining from the nose after surgery (possible CSF leak) with headache or fever.
- Rapidly worsening symptoms of any kind.

> Pituitary apoplexy and adrenal crisis are life-threatening emergencies. If you carry a steroid emergency kit or alert card, use it and call emergency services immediately.

17. Follow-Up Schedule

Follow-up watches hormone levels, vision, and the tumor. It is often lifelong because hormone balance needs ongoing care.

- Initial follow-up: within a few weeks of surgery to check healing, vision, and hormones.
- MRI scans: a baseline scan after treatment, then periodic scans (often at 3–12 months and then yearly, spacing out if stable).
- Hormone testing: regular blood tests to check for over-production, under-production, and to adjust replacement.
- Vision testing: repeated visual field checks if the tumor pressed on the optic nerves.
- Neurological examination: as needed at reviews.
- Long-term monitoring: often lifelong for hormone balance and recurrence.
- Recurrence surveillance: new symptoms or rising hormone levels prompt earlier review.

18. Long-Term Outlook

The outlook for most pituitary tumors is very good. Many are cured or well controlled, and vision and hormone problems often improve with treatment.

- Prognosis: excellent for most benign adenomas; prolactinomas are often controlled with medicine alone.
- Quality of life: usually good, especially once hormones are balanced and vision recovers.

- Recurrence risk: possible, so periodic scans and hormone tests continue.
- Living with the condition: many people need lifelong hormone replacement or monitoring, which becomes a manageable routine.
- Psychological support: helpful for coping with body-image changes and the demands of long-term hormone care.

Your endocrinologist and neurosurgeon can give the most accurate outlook based on the tumor type, size, and treatment response.

19. Prevention and Risk Reduction

There is no proven way to prevent pituitary tumors, but general steps support health and help detect problems early.

- Healthy lifestyle: balanced diet, regular activity, healthy weight, and good sleep.
- Report symptoms early: vision changes, persistent headaches, menstrual changes, or unexplained body changes deserve medical review.
- Manage chronic conditions: control blood pressure and blood sugar, which hormone tumors can worsen.
- Regular medical check-ups: especially if you have a family history or a known syndrome such as MEN1.
- Screening: routine screening is not done in the general population; people with inherited syndromes may have regular monitoring.
- What cannot be prevented: most pituitary tumors arise without an avoidable cause.

20. Frequently Asked Questions (FAQ)

Below are common questions from patients and caregivers about pituitary tumors. Answers are general; your endocrinologist and surgeon can advise for your situation.

Q1. What is a pituitary tumor?

It is a growth in the pituitary gland, the small hormone-control gland at the base of the brain. Most are benign adenomas that are slow growing.

Q2. Is a pituitary tumor cancer?

Almost never. The vast majority are benign (non-cancerous) adenomas. True pituitary cancer (carcinoma) is very rare.

Q3. Can a pituitary tumor spread?

Benign adenomas do not spread to other organs. They can grow and press on nearby structures such as the vision nerves. Spread is only seen in the very rare carcinoma.

Q4. Will I need surgery?

Not always. Prolactinomas are often treated with medicine alone. Surgery (usually through the nose) is used for tumors pressing on vision or making certain hormones.

Q5. What is transsphenoidal surgery?

It is an operation to remove the tumor through the nose and sphenoid sinus, avoiding opening the skull. Recovery is usually quicker than a craniotomy.

Q6. Can a pituitary tumor come back?

Yes, it can regrow, so long-term monitoring with scans and hormone tests continues even after successful treatment.

Q7. Is a pituitary tumor hereditary?

Usually not. Most occur by chance. A few are linked to inherited syndromes such as MEN1; tell your doctor of any family history.

Q8. Why do I have vision problems?

A larger tumor can press on the optic chiasm just above the gland, classically causing loss of side vision. Relieving the pressure often improves vision.

Q9. Can I drive?

Driving is usually paused after surgery or if your vision is affected. You can return after clearance and normal vision testing, per your region's rules.

Q10. Can I travel?

Yes, once stable and cleared. If you take steroid replacement, carry extra medicine, a steroid alert card, and know your sick-day rules.

Q11. Can I fly?

Flying is generally safe once cleared, but avoid it soon after transsphenoidal surgery until your surgeon agrees, to protect the healing nasal area.

Q12. Can I work?

Most people return to work within a few weeks after surgery for desk jobs. Timing depends on your recovery, vision, and hormone balance.

Q13. Can I exercise?

Gentle activity is encouraged, but avoid heavy lifting, straining, and forceful nose-blowing for a few weeks after nasal surgery. Ask your surgeon when to resume.

Q14. Can I have children?

Often yes. Treating a prolactinoma can actually restore fertility. Pregnancy needs planning and monitoring, especially with larger tumors — discuss it with your endocrinologist.

Q15. Can I drink alcohol?

Limit alcohol, especially during recovery and if you take medicines that interact with it. Ask your doctor about your specific treatment.

Q16. Is an MRI safe?

Yes. MRI uses magnets, not radiation, and a dedicated pituitary MRI is the best test for the gland. Tell staff about metal implants or a pacemaker.

Q17. Will I lose my memory?

Memory loss is not typical. Hormone imbalance can affect concentration and mood, which usually improve once hormones are balanced.

Q18. What symptoms require emergency care?

A sudden severe headache with vision loss or vomiting (possible apoplexy), sudden vision loss, collapse, or signs of adrenal crisis on steroid replacement — seek emergency care at once.

Q19. How often will I need follow-up?

Often a scan and hormone tests at 3–12 months, then yearly if stable, with vision checks if the tumor pressed on the nerves. Hormone care is frequently lifelong.

Q20. What is a prolactinoma?

A common pituitary tumor that makes too much prolactin, causing irregular periods, milk production, low libido, or infertility. It is usually treated with tablets that shrink it.

Q21. What is acromegaly?

A condition from too much growth hormone, causing enlarged hands, feet, and facial features and joint pain. It is treated with surgery, medicine, or radiation.

Q22. What is Cushing's disease?

A condition from a tumor making too much ACTH, raising cortisol. It causes weight gain, high blood pressure, and high sugar, and is usually treated with surgery.

Q23. What does 'functioning' vs 'non-functioning' mean?

A functioning tumor over-produces a hormone and causes specific symptoms; a non-functioning tumor does not and usually causes problems by its size.

Q24. What is the difference between a microadenoma and a macroadenoma?

A microadenoma is under 10 mm; a macroadenoma is 10 mm or larger and is more likely to press on the vision nerves.

Q25. Why do I need hormone replacement?

If the tumor or its treatment leaves the gland underactive, you may need to replace missing hormones such as thyroid hormone, steroids, or sex hormones to feel well and stay safe.

Q26. Why must I never stop steroid replacement suddenly?

Stopping cortisol replacement abruptly can cause a life-threatening adrenal crisis. Follow sick-day rules and always keep an emergency supply and alert card.

Q27. What is diabetes insipidus?

A temporary (sometimes lasting) problem after pituitary surgery causing very frequent urination and intense thirst. It is treated with a medicine that replaces the missing hormone.

Q28. Will radiation affect my hormones?

It can, over time, reduce pituitary function, so your hormones are monitored for years after radiation and replaced if needed.

Q29. Is chemotherapy used?

Not for typical adenomas. A specific drug (temozolomide) is reserved for rare aggressive tumors, under specialist care.

Q30. Is a pituitary tumor painful?

It can cause headaches from stretching nearby tissues, but the gland itself does not feel pain. Headache is managed as part of care.

Q31. Is it contagious?

No. You cannot catch or pass on a pituitary tumor.

Q32. Will my appearance change?

Hormone excess (acromegaly or Cushing's) can change appearance; many changes stabilise or partly improve after treatment. Support and counselling help.

Q33. Can children get pituitary tumors?

Yes, though less commonly. They can affect growth and puberty, so children are cared for by paediatric endocrine specialists.

Q34. How long is the hospital stay?

Often about 2–4 days after transsphenoidal surgery if recovery is smooth.

Q35. Can I blow my nose after surgery?

Not forcefully for the first few weeks after nasal surgery, to protect the healing area. Sneeze with your mouth open and follow your surgeon's advice.

Q36. Can I take supplements?

Check with your doctor, as some can interact with hormone treatment. A balanced diet is usually the best source of nutrients.

Q37. Will I need lifelong treatment?

Some people do — either ongoing medicine (as for many prolactinomas) or hormone replacement and monitoring. This becomes a manageable routine.

Q38. How is the tumor type confirmed?

By combining MRI with detailed hormone blood tests, and by examining tissue removed at surgery.

Q39. How do I cope emotionally?

Hormone changes and diagnosis can affect mood. Counselling, support groups, and open communication with your endocrine team all help.

Q40. Does diet cure a pituitary tumor?

No diet cures it, but good nutrition supports recovery and helps manage weight and blood sugar, which some hormone tumors affect.

Q41. Who is on my care team?

Usually an endocrinologist, a neurosurgeon (often a pituitary specialist), an ophthalmologist for vision, a radiologist, nurses, and sometimes a radiation oncologist.

21. Metadata for Retrieval

Structured tags to help the AI assistant retrieve this document accurately.

Field	Value
Tumor Name	Pituitary Tumor
Alternative Names	Pituitary adenoma, prolactinoma, growth-hormone/ACTH-secreting adenoma, non-functioning adenoma
WHO Grade	Usually benign adenoma (not WHO I–IV graded); rare pituitary carcinoma
Tumor Category	Tumor of the pituitary gland (endocrine/skull-base)
Brain Region	Pituitary gland in the sella turcica at the base of the brain, below the optic chiasm
Benign or Malignant	Almost always benign; malignant carcinoma is very rare
Typical Symptoms	Vision loss (side fields), headache, hormone excess (prolactin, growth hormone, cortisol) or deficiency, fatigue, menstrual changes
Diagnostic Methods	Pituitary MRI, contrast MRI, hormone blood tests, visual field testing, genetic testing if syndrome suspected
MRI Characteristics	Sellar mass; micro (<10 mm) or macroadenoma (>=10 mm); suprasellar extension compressing optic chiasm; possible apoplexy
Treatment Options	Observation, transsphenoidal/endoscopic surgery, radiosurgery/radiation, hormone therapy (dopamine agonists, somatostatin analogues), hormone replacement
Recovery Duration	Transsphenoidal surgery 4–6 weeks; medicine-controlled tumors improve over weeks to months
Follow-Up Frequency	MRI at 3–12 months then yearly; regular hormone and vision testing, often lifelong
Common Complications	Vision loss, hypopituitarism, diabetes insipidus, CSF leak, apoplexy, recurrence, mood changes
Emergency Symptoms	Sudden severe headache with vision loss (apoplexy), adrenal crisis on steroids, CSF leak, collapse
Keywords	pituitary tumor, pituitary adenoma, prolactinoma, acromegaly, Cushing's disease, transsphenoidal surgery, hormone replacement
Synonyms	Pituitary gland tumor, hypophyseal adenoma, sellar tumor
Related Medical Conditions	Hypopituitarism, diabetes insipidus, acromegaly, Cushing's disease, MEN1, visual field loss